THE NEW ROBOT BRAIN IN ONE PAGE

Our Robot Can Think in Small Steps

It plans a safe trip, follows the path, checks the camera, shoots, and parks - one job at a time.

Think of the code like a careful helper with a checklist. It finishes one small job before starting the next one.

1

LOOK
The Limelight camera looks for the Hive target that matches our color and field side.

2

PLAN
A map helper draws a safe route around the Hive instead of guessing where to drive.

3

DRIVE
Pure Pursuit follows the route. If the robot gets bumped or the camera fixes its position, it can plan again.

4

CHECK AND SHOOT
The shooter starts after the robot begins moving. The feeder waits until speed and camera range are both ready.

5

PARK
When the shot is finished - or time gets short - the robot heads to a safe parking place.

What changed behind the scenes

SEPARATE HELPERS	PLANS CAN CHANGE	BUILT IN STOP SIGNS
Driving, camera, shooter, intake, and transfer each have a smaller job.	A route can be rebuilt after a bump or a trusted camera correction.	Timeouts, current checks, failure states, and emergency parking add protection.

The driver's recipe

1	2	3	4
Pick red or blue	Pick audience or far side	Hold the assist button	Let go to take over

Safety lock before robot testing

The new modes stay disabled until the team measures the field, checks motor directions and speeds, tests sensors, calibrates the Limelight position, and confirms the shooting and parking spots.